UNIVERSITI TEKNOLOGI MALAYSIA

MID- TERM EXAMINATION

SEMESTER II 2014/2015

SUBJECT CODE	**: SCSI 2143**
SUBJECT NAME	**:PROBABILITY & STATISTICAL DATA ANALYSIS**
YEAR/COURSE	**: 2, 3 /SCSB, SCSJ, SCSR, SCSV**
TIME	**: 2 HOURS**
DATE	**: 14TH APRIL 2015**
VENUE	**: BK 1–5, N28**

INSTRUCTIONS TO THE STUDENTS:

Answer all questions in the given answer booklet.

Name	
I/C No.	
Year/Course	
Section	
Lecturer Name	

This questions paper consists of **6** printed pages including this page.

QUESTION 1 (10 MARKS)

i) A survey was conducted by a government agency to get the information on the number of days dengue fever was suffered by infected victim in Malaysia before blood test was done. This has led to following information: 40% get a blood test done after more than 4 days of having fever. 35% get a blood test done after 3 to 4 days of having fever. Another 25% get a blood test done after 1 to 2 days of having fever. (Note: Data is arbitrary)

a) Are the given percentage values computed from population or sample? (1 mark)

b) State whether the data collected is discrete or continuous? Justify your answer. (2 marks)

c) Base on the given data above, draw the best graphical representation to present the data meaningfully. (3 marks)

ii) The level of measurement refers to the relationship among the values that are assigned to the attributes for a variable. There are four level of measurements can be used.

a) State any two of the level of measurements with order as one of the properties. (2 marks)

b) The following question in Figure 1 is extracted from a survey. What is the level of measurement used in this question? Explain your answer. (2 marks)



Figure 1: A sample question extracted from a survey

QUESTION 2 (10 MARKS)

Fifteen (15) people were asked to state the number of hours they exercise in a seven days period. The results of the survey are listed below.

8, 2, 4, 7.5, 10, 11, 5, 6, 8, 12, 11, 9, 6.5, 10.5, 13

i) Complete the frequency table below (Table 1). (5 marks)

Table 1

Hours of Exercise	**Tally**	**Frequency**
0 – 2		
3 – 5		
6 – 8		
9 – 11		
12 – 14		

ii) Plot a histogram to display the data. (3 marks)

iii) Make some conclusions based on the information displayed in the histogram.

(2 marks)

QUESTION 3 (12 MARKS)

i) The value of 11 houses in a new development are shown in the table below (Table 2):

Table 2

Value per House	Number of Houses
RM 100,000	1
RM 175,000	5
RM200,000	4
RM700,000	1

a) Find the mean value of these houses in RM. (2 marks)

b) Find the median value of these houses in RM. (2 marks)

c) State which measure of central tendency, the mean or the median, "**best**" represents the values of these 11 houses. Justify your answer. (2 marks)

ii) The length of stay in the first class ward of Hospital Kenanga was organized into a frequency distribution. The mean length of stay was 28 days, the median 25 days and mode length is 23 days. The standard deviation was computed to be 4.2 days.

a) Is the distribution symmetrical or skewed? Justify your answer. (2 marks)

b) Based on the given measurement of central tendency, determine whether the skewness is positive, negative or zero. Sketch the appropriate graph to show the dispersion of the central tendency. (4 marks)

QUESTION 4 (5 MARKS)

The probability that a new airport will get an award for its design is 0.16, the probability that it will get an award for the efficient use of materials is 0.24, and the probability that it will get both awards is 0.11.

i) What is the probability that it will get at least one of the two awards? (3 marks)

ii) What is the probability that it will not get any awards? (2 marks)

QUESTION 5 (23 MARKS)

i) Suppose that you take a five-question multiple-choice quiz by guessing. Each question has possible answers *a*, *b*, *c*, *d* and only one is correct. Let *x* denote the number of randomly selected question have correct guessing

a) What is the probability distribution of *x*? (1 mark)

b) State the value of *n* and *p*. (1 mark)

c) Calculate the mean of this distribution. (1 mark)

d) What is probability that you guess exactly one answer correctly? (1.5 marks)

e) What is the probability that you guess more than two answers correctly? (2.5 marks)

ii) A basketball player makes 80% of his free throws. We put him on the free-throw line and ask him to shoot free throws until he misses. Let *x* = the number of free throws the player takes until he misses.

a) What is the probability distribution of *x*? (1 mark)

b) What is the probability that the player misses his first free throw within the first 3 attempts? (2 marks)

c) What is the probability that the player shoots at least 4 free throws before misses his first throw? (4 marks)

iii) Ahmad and Syahmi belong to an athletics club. In javelin throwing competitions, Ahmad throws is normally distributed with mean 41.0 m and standard deviation 2.0 m. While, 85% of Syahmi's javelin throws exceed 35m and 70% exceed 37.5 m. Syahmi's throws is also normally distributed, with standard deviation, σ = 6.75.

a) What is the probability of Ahmad's throws between 40 m and 46 m? (2 marks)

b) Find the mean of Syahmi's throws, in two decimal places. (3 marks)

c) In order to qualify for the final round, it is necessary to achieve a throw of at least 48 m in the preliminary rounds. (3 marks)

I. Calculate the probability of Ahmad qualify for the final round.

II. Calculate the probability of Syahmi qualify for the final round.

d) Based on results in (c), who should be chosen, Ahmad or Syahmi? Justify your answer. (1 mark)

Formulas

$$mean, \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \qquad \boldsymbol{variance}, \boldsymbol{s^2} = \frac{\sum_{i=1}^{n} (\boldsymbol{x_i} - \bar{\boldsymbol{x}})^2}{\boldsymbol{n-1}}$$

$$\boldsymbol{P(x) = \frac{n!}{x!\,(n-x)!} p^x (1-p)^{n-x}} \qquad \boldsymbol{P(x) = (1-p)^{x-1} p}$$

$$Z = \frac{X - \mu}{\sigma}$$